Hiding the peak, not missing the days: support-based screening of incomplete river-temperature archives

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Key points

- A gap of fixed length damages the hottest 30-day mean only when it hides that window; the annual mean reacts to length
- At a 5% error cap, a rule that screens hidden peak support keeps 67% of an external archive; a gap-length rule keeps 40%
- A fitted risk score adds 7–10 points over the support rule, but real calendars may need no screen and refusal costs hot years

Abstract

Incomplete river-temperature records are admitted by counting missing days: a year passes if its longest gap is short enough. A gap of a given length does not remove the same amount of the statistic computed, and in real records the two vary together, so they have never been separated. We pair 5,114 complete station-years from 146 river networks with 53,721 designed outages that vary gap length and hidden support independently, and with 418 real monitoring calendars, and score admission of the annual hottest 30-day mean. Hidden support, not gap length, orders the error: a 30-day gap that hides the peak window costs 0.68 °C, the same gap outside it 0.03 °C, while the annual mean responds to length, not position. The reconstructor sets the height; residual error concentrates in cases whose reconstructed window lands away from the true one. At a 5% cap on the error of the admitted reconstruction, support screening keeps 67% of a 13,017-case external panel and a gap-length rule 40%, while a fitted risk score keeps 74%, 7 to 8 points above the support rule and not reliably better than a calibrated bound. On 418 real calendars, which hide less than half of the peak window in 94% of cases, admitting everything sits at 2.6–4.1%, under the cap, and screening refuses a quarter to a half of the true hot years. Support is the right conditioning variable, but its value is confined to archives whose gaps are consequential.

Plain Language Summary

Rivers are warming, and managers ask how hot a river got in its hottest month. Answering needs a year of daily temperatures; records have holes; and the usual screening test asks how many days are missing. We show that this is the wrong question for a summer peak. What matters is whether the missing days include the few weeks the peak depends on. Using complete years from 146 river networks, we place gaps of the same length in different positions and compare what two statistics lose: a gap over the peak window destroys the hottest 30-day mean while the same gap in autumn barely moves it, and a whole-year average behaves in the opposite way. Screening on this hidden support therefore keeps two thirds of a large test archive where a rule based on gap length keeps two fifths. A learned risk score keeps slightly more again, but it is complex, it does not beat a simpler calibrated alternative, and on the holes that real monitoring programmes actually leave the peak is usually observed anyway, so no screen is needed. Screening buys

usable years only where gaps are long and badly placed, and it buys them by discarding hot years.

1 Introduction

Cold-water habitat assessment, thermal-regime classification and criterion-exceedance accounting (Washington State Department of Ecology, 2024) all reduce a year of daily water temperature to one number — a hottest 30-day mean, an annual mean, a seasonal amplitude (Caissie, 2006; Isaak et al., 2017) — and it is that number, not the daily series, that enters a management judgement. The error that matters is therefore the error of the derived statistic. On the designed outages placed on the complete station-years assembled here, computing the summer peak from the recorded days alone, which is what an agency does once a year passes a completeness test, carries a mean absolute error of 1.31 °C, and 24.9% of station-years are then wrong by more than 1.0 °C. An error of that size turns a thermally stressful year into an unremarkable one.

Current practice assumes that the amount of missing data decides whether the derived number may be used, and two families of rule encode that assumption (Table 1). Collection standards specify what must be present: Mauger et al. (2015), writing minimum standards for a state whose records feed regional analyses, require a minimum sampling duration of one calendar month and recommend a deployment period of 1 June to 31 August "to capture the thermal maxima"; the same standards govern which of those records enter the regional assessment (Mauger et al., 2017). Quantitative guidance constrains what may be absent: Johnson et al. (2021) report that "datasets with less than seven weeks of consecutively missing data enabled estimation of all annual signal parameters with reasonable accuracy", expanding to "approximately 20 weeks" when missing data are imputed. The gap-filling literature optimises the accuracy of the filled series rather than the admissibility of the completed year (Mohseni et al., 1998; Li et al., 2017), and treats the incomplete record as a missing-data problem rather than as a property of the derived statistic (Rubin, 1976), and the closest prior work fills incomplete summer records so that a thermal peak can be computed, applies that filling uniformly, declines no station-year, and admits July–August-only records on the climatological position of the hottest window (Beaufort et al., 2022). Large-sample thermal models

optimise daily or reach-scale prediction skill (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023), and selective-prediction methods control error at a fixed rejection rate (Geifman & El-Yaniv, 2017; Denis et al., 2020), and distribution-free risk control bounds an expectation under an exchangeability assumption (Lei et al., 2018; Angelopoulos et al., 2022; Bai et al., 2026). None of these rules measures how much of the target statistic's own support a record contains.

The assumption is plausible for some statistics and wrong for others. An annual mean is a sum over days, so removing k days removes k terms and it matters little which. A hottest 30-day mean is an extreme supported on a short window: if the outage misses that window the statistic is recoverable almost exactly however long the outage is, and if the outage covers it the statistic must be inferred however short the outage is. The direction of a completeness rule's error should therefore be opposite in the two cases, and an average over a mixed archive would conceal the reversal. The literature is consistent with that reversal without having isolated it: reported timing effects on annual thermal parameters are inconsistent across regions, years and parameters (Johnson et al., 2021); reconstruction bias depends on where the target window falls (Beaufort et al., 2022); paired air–water analyses show how much of the thermal regime is inherited from the atmosphere (Hare et al., 2023); large-scale climate modes modulate river thermal regimes (Sanikhani et al., 2026; You et al., 2026); systematic, time-varying instrument bias rather than random noise dominates long-term monitoring error (Jackson et al., 2025); and statistic-aware completeness conventions do exist in climate monitoring, but they are defined by counts or seasons rather than by the statistic's support (Zhang et al., 2011). Irregular sampling and sampling artifacts likewise bias means and trends and decide which model structures remain trustworthy (Grey et al., 2024), and reconstruction frameworks have been built for regulated and data-sparse basins, including Chinese catchments and the Yangtze mainstem (Huang et al., 2023; Wang, 2025); those studies answer estimation and reconstruction questions, not the archive-admission question for a named annual extreme.

Testing the assumption requires separating two properties of a gap that real records entangle. Gap length and the share of the target window hidden correlate at $+0.45$ in our own panel, and gap length correlates at -0.86 with the fraction of warm-season days that remains observed; a comparison across real gaps therefore

cannot say whether a rule works because it avoids long gaps, because it avoids badly placed ones, or because both simply track how much warm-season record the station has. A second obstacle is truth: in exactly the years where the admission decision matters, the target statistic is unknown by construction, so the benefit of a rule cannot be measured where it would be applied. A third is that evaluations report a rule's discrimination rather than the composition of the archive it leaves behind, and composition is what decides which catchments remain available for analysis.

To address these limitations we assemble 5,114 complete river station-years from 146 networks and nine national agencies, place 53,721 outages on complete years so that the two properties of a gap can be varied independently, and score admission rules on held-out river networks, on an external panel that took no part in development, and — without any truth — on the 2,422 records in this same archive that really do have gaps. Three questions organise the paper. First, which property of a gap predicts the error of a derived thermal statistic, and does the answer depend on which statistic is being computed? Second, how much of that dependence is the reconstructor's doing, and what mechanism explains the error that remains when support is accounted for? Third, what does conditioning on support change for the error among retained station-years, for which stations survive in the archive, and for the hot-year judgement that the archive is used to make? The third question decides whether the first two have practical value, and on the evidence below they have value only in a regime we can name in advance.

2 Study system, data and design

2.1 The statistic and its support

Let a station-year be a sequence $y = (y_1, \ldots, y_T)$ of daily mean water temperatures and let M be the set of missing days. The annual hottest 30-day mean is

$$H_{30}(y) = \max_{t \in \{30,\ldots,T\}} \frac{1}{30} \sum_{s=t-29}^{t} y_s, \tag{1}$$

and its support is the set of days inside the winning window, $S = \{s : s \in [t^* - 29, t^*]\}$. For a gap M we distinguish two quantities that real records confound:

$$\ell = \max\{\text{length of a consecutive run in } M\}, \qquad h = \frac{|M \cap S|}{|S|}. \tag{2}$$

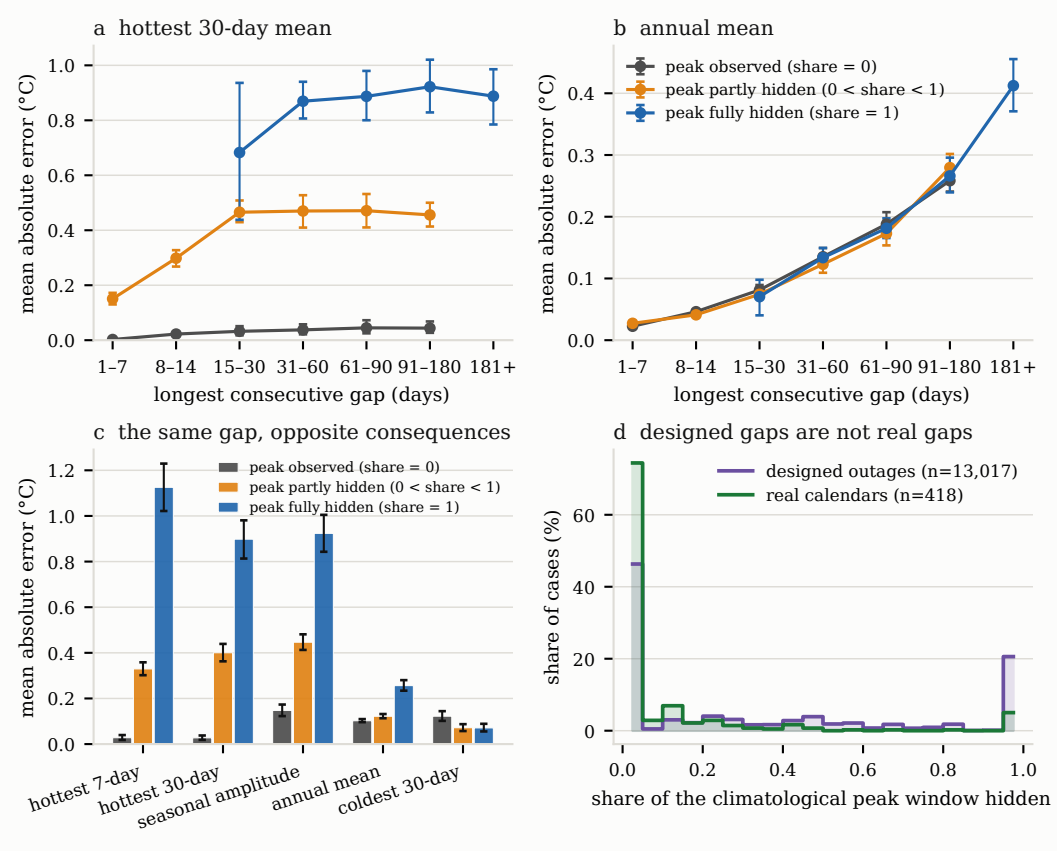


Figure 1. Hidden support, not gap length, orders the error, and the order reverses between statistics. All panels use the 13,017 designed outages of the external panel with a day-of-year climatology reconstruction, so the panels differ only in the statistic and in gap geometry. (a) Mean absolute error of the hottest 30-day mean against the longest consecutive gap, split by the share of the true 30-day peak window that the gap hides (grey, none; orange, partial; blue, all). Errors are network-clustered means with 95% intervals (2,000 cluster-bootstrap draws over the 41 shared-station groups); a cell is drawn when it holds at least 25 cases. (b) The same for the annual mean. (c) Mean absolute error of five annual statistics at the three hidden-support classes, with the same intervals; the coldest 30-day mean moves in the opposite direction to the hottest. (d) Distribution of hidden peak support in the designed outages (purple) and in the 418 real monitoring calendars (green). Source data: `fig1_source.csv`.

Table 1. What each body of work optimises, decides on, and conditions on. The rows are ordered by what they assume about the conditioning variable, not by date.

body of work	optimises	decision unit	conditioning axis
completeness guidance (Johnson et al., 2021)	tolerable amount of missing data	station-year	outage size (timing tested; effects reported inconsistent)
collection standards (Mauger et al., 2015)	comparability across a programme	deployment	season and duration
gap filling (Mohseni et al., 1998; Li et al., 2017)	accuracy of the filled series	day	air temperature, donors
peak reconstruction (Beaufort et al., 2022)	spatial coverage of the thermal peak	reach	air regression applied uniformly; climatological summer window used to admit partial-season records
large-sample models (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023)	prediction skill	day or reach	meteorology, catchment attributes
selective prediction (Geifman & El-Yaniv, 2017; Denis et al., 2020)	error at a fixed rejection rate	case	model confidence
this work	**error of the derived statistic among retained years**	**station-year**	**share of the statistic's support hidden**

Here ℓ is what a completeness rule reads and h is the share of the statistic's own support that the record does not contain.

Two properties of this definition matter for what follows. First, S is defined by the year's own temperatures and is therefore unknown at the moment of the admission decision. The admission rules below use an observable analogue, $\hat{h}$, computed against the station's climatological peak window estimated from its other complete years; the diagnostic panels use the true h, and the difference between the two is one of the mechanisms we report. Second, h is statistic-specific: for the annual mean every day is support, so h degenerates to the missing fraction, whereas for the coldest 30-day mean the support is the winter window, and a gap that hides the summer peak is a gap that leaves the cold support observed. The same gap therefore has three different values of h and can be good for one statistic and bad for another.

2.2 Records and panels

Daily mean water temperature comes from nine national providers (ARSO, CHMI, FOEN, GKD Bayern, LUBW, NVE, Rijkswaterstaat, SYKE and USGS) and, on the external panel, from France's Hub'Eau and the UK Environment Agency. All records are public monitoring archives reused for this analysis; no sampling was conducted for this study. The base panel keeps networks with at least four complete years: 146 networks, 457 stations and 5,114 fully observed station-years. Two derived panels are used. The development panel holds 110 networks, 354 stations, 4,045 station-years and 40,704 designed outages; it was used to select methods and is not a confirmation set. The external panel holds 49 network labels, 143 distinct stations, 1,103 station-years and 13,017 designed outages. Labels that share a station are merged into connected components before any fold is drawn, so the external panel has 41 validation groups and no target station appears in two folds; 164 station-years appear under more than one label and stay inside one group.

The stored policy reconstruction fails the 0.5 °C tolerance on 15.3% of external cases and 14.9% of development cases; those are the rates a screen has to improve on. The real, incomplete part of the archive is 2,422 station-years. Those years have no complete counterpart, so no reconstruction error can be measured on them; they

are used only to describe the gap geometry that monitoring programmes actually leave.

2.3 Designed outages: varying length and support independently

On each complete station-year we place synthetic outages with a factorial design: a target gap length in {30, 60, 90, 180} days and a position drawn so that the gap falls in one of three support classes — clear of the peak window ($h = 0$), partly overlapping it ($0 < h < 1$), or covering it completely ($h = 1$) — with the remaining missing days scattered at random to reach the designed completeness. This is what makes the two properties separable: h is set by the position draw and ℓ by the length draw, and their correlation in the designed panel falls to the design value instead of the +0.45 observed in real records. Each outage is reconstructed by five methods that see the same information (Table 2).

2.4 Real calendars

The 418 candidate groups of the real-calendar replay are built by transplanting every missing day of a genuinely incomplete year onto a complete target year, keeping the calendar position of the gap. This is the only setting in which the gap geometry is the one an agency actually faces. It has no new sealed panel and it is not a confirmation set; it is a development-period probe of what screening would do to real calendars.

3 Methods

3.1 Reconstruction and the information it may use

Every reconstructor is fitted leave-one-year-out: the model that fills days in year y sees only other years at the same station, plus, where the method allows it, other stations of the same network. Five reconstructors are used: day-of-year climatology, a smoothed climatology, linear interpolation between observed days, a ridge regression on visible donor stations, and a ridge regression on gridded air temperature (NASA POWER (NASA Langley Research Center, 2026)). A stored per-case policy selects one of them for each station-year on validation evidence, and the reconstruction that the admission rule accepts is that policy output, whose

Table 2. Panels, their units, and what each is allowed to support. "Groups" are connected components of networks that share a station id and are the resampling and fold unit throughout.

panel	cases	networks	groups	role
complete base	5,114 station-years	146	—	reference truth for every recovery experiment; USGS reference-network criteria follow (Falcone, 2011)
development	40,704 designed outages	110	110	method selection and cross-fitted development evidence
external	13,017 designed outages	49 labels, 143 stations	41	cross-fitted external comparison; already opened, not a new seal
real archive	2,422 incomplete station-years	—	—	gap geometry only; no complete counterpart, so no error can be measured
real calendar replay	418 candidate groups	14	14	development-period probe of deployment on real gap calendars

absolute error we write e and whose consequential failure we define as $e > 0.5\,^{\circ}\mathrm{C}$. On the external panel the policy chooses donor regression for 69% of cases, smoothed climatology for 24% and interpolation for 6%; the climatology fallback is where the policy fails, at 38% against 8% for donor regression, and it is one reason the fitted score can separate records that share the same hidden support. The tolerance is fixed before any result in this paper was computed; it is roughly half the interannual standard deviation of the hottest 30-day mean within a station, so a failure is a year whose reconstructed rank would move.

3.2 The admission problem

A rule computes a score $s(x)$ from decision-time features x and admits a record when $s(x) \le \tau$. The threshold τ is selected on calibration groups only, never on the labels of the records it will judge. We compare four scores that are available before the decision and one reference arm:

- *gap-length rule*: $s = \ell$, the longest consecutive missing run.
- *peak-window rule*: $s = \hat{h}$, the share of the climatological peak window hidden by the gap.
- *certificate bound*: $s = b$, a calibrated upper bound on the reconstruction error stored by a companion certificate model; it ranks records by predicted error and carries no label from the evaluated year.
- *fitted risk score*: a quantile gradient-boosting regressor ($q = 0.9$, depth 4, 350 iterations) on 22 decision-time features that include gap geometry, visible-donor count and correlation, the reconstructor's own training residual and the disagreement among reconstructors. It is trained on e, the error of the same policy reconstruction it is used to admit, and it is cross-fitted by group: the calibration score of each outer fold is an inner out-of-fold prediction, so no outer test label can reach that fold's threshold.
- *reconstruct everything*: the reference arm, admitting every record whose policy reconstruction is defined. It is what an agency does if no screen is worth its cost, and it is the arm against which the value of screening has to be judged.

Ties are handled once, at the score definition: a threshold is an observed value of the raw score, and admission is $s \le \tau$, so a tie block is admitted whole in calibra-

tion, in evaluation and at deployment. No jitter is added at any stage. A separate fixed-retention comparison admits the safest 70% of each fold using a deterministic case key to break ties, which is a second deployable rule and is reported as such.

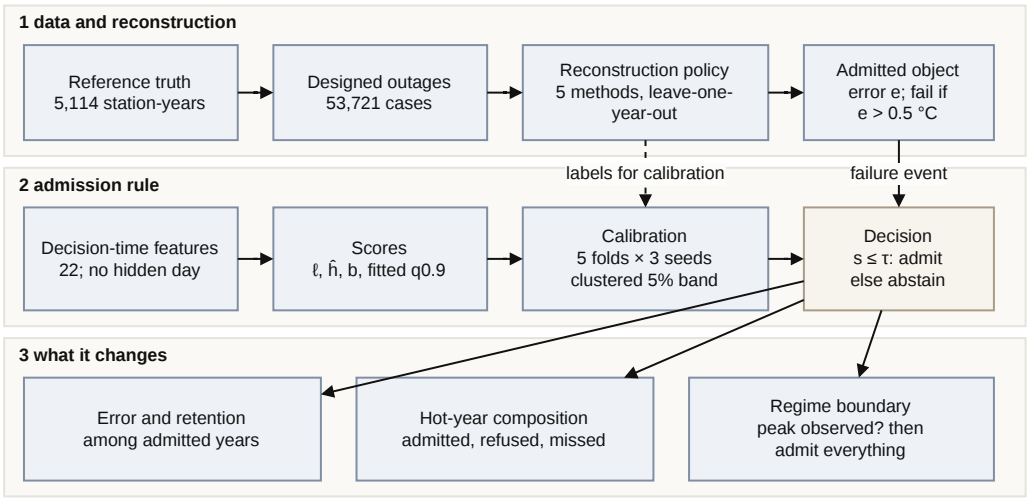


Figure 2. How a station-year becomes an admissible record. Band 1: complete station-years supply the reference truth, designed outages vary gap geometry, five reconstructors are fitted leave-one-year-out, and the stored policy output is the admitted object, whose error against the complete year defines a failure at 0.5 °C. Band 2: the admission rule reads only decision-time information, scores each record, calibrates one threshold per group fold with a clustered bound at the 5% cap, and admits a record whose score is at or below the threshold; ties are admitted whole. Band 3: what the rule changes, and the regime check that decides whether a screen is worth its cost. Schematic source: `f2_system_design.drawio` (draw.io; see the Open Research statement).

3.3 Risk control and what it means

The threshold is the largest candidate whose group-clustered bootstrap bound on the calibration false-admit rate is at most 5%, with the percentile taken at $1 - 0.05/n_{\text{cand}}$ over the n_{cand} distinct thresholds searched. We are explicit about what this is and is not. It is an empirical, finite-sample calibration rule for a clustered design, in the spirit of distribution-free risk control but without its exchangeability guarantee (Angelopoulos et al., 2022; Bai et al., 2026). It is not a distribution-free guarantee for an unseen archive, and a bound that holds at each candidate is not a

bound that holds after maximising over candidates; the Bonferroni factor is there for that reason. Section 4.5 reports what the rule actually delivers in simulation, separating the population risk of the selected threshold from the error rate observed in a finite test batch — the two are not the same event, and only the first is a property an agency can rely on.

3.4 Uncertainty and the unit of inference

The resampling unit is the shared-station group. All intervals are percentile intervals from 2,000 group-bootstrap draws unless stated otherwise, and every contrast is paired: the same resampled groups are used for both arms. Three fold seeds (101, 202, 303) are pre-declared and all are reported. Shares are fractions throughout; we write percentage points when a difference is in points. For the hot-year analysis, the case-weighted estimate is primary and a unique station-year weighting, in which each station-year carries total weight one regardless of how many outage realisations it contributes, is reported as the pre-declared sensitivity.

4 Results

4.1 The same gap has opposite consequences for different statistics

4.1.1 Support orders the error, and length barely moves it

Figure 1a and b hold the gap length in view and split by hidden support. For the hottest 30-day mean the support class sets the height of the curve and the length barely moves it: a gap that leaves the peak window observed costs 0.003 °C at one week and 0.04 °C at 91–180 days, while a gap that covers the window costs 0.68 °C at 15–30 days and stays near 0.9 °C from 60 days out to beyond 180. At a fixed length of 30 days the crossed design puts the position effect at 0.64 °C [0.55, 0.74], and at 90 days at 0.89 °C [0.76, 1.04]. For the annual mean the same split collapses: the three support classes lie on top of one another and the error rises with length from 0.02 °C at one week to 0.26 °C by 180 days, with the one class that has longer records reaching 0.41 °C.

4.1.2 A colder statistic reverses the ordering

The reversal is not confined to the annual mean. Figure 1c compares five annual statistics at the three support classes. The hottest 7-day mean behaves like the hottest 30-day mean but more steeply (0.03 against 1.13 °C). The seasonal amplitude, which is a difference of a summer and a winter window, behaves like the hot statistics because its summer term is the one at risk. The coldest 30-day mean moves the other way: 0.12 °C when the gap misses the peak window against 0.07 °C when it covers it, because a gap placed away from the summer peak is a gap that can cover the winter minimum, and a gap over the summer peak leaves the cold support observed. One rule, three statistics, opposite consequences: what a completeness count cannot express is which days a statistic is made of.

Figure 1d adds the reason this matters for practice. In the designed panel 30% of outages hide more than half of the climatological peak window and the median gap is 30 days. In the 418 real calendars the median gap is 7 days, the median missing fraction is 2.2%, and 5.7% of candidates hide more than half of that window. The two populations are not the same population, and Section 4.4 shows what that difference does to the value of a screen.

4.2 The reconstructor sets the height, not the shape

The support effect is conditional on what the reconstructor can infer. Figure 3a crosses gap position with gap length at 30, 60 and 90 days for five reconstructors. Hiding the peak always costs something, but the cost ranges from 0.24 °C [0.20, 0.28] for donor-station regression to 0.64 °C [0.55, 0.74] for day-of-year climatology at 30 days, and the ordering of the five methods at 90 days is the same as at 30.

Figure 3b asks the complementary question: with the hidden share held at one, does length still matter? For climatology and smoothing the answer is a few hundredths of a degree and not distinguishable from zero under a network-paired interval (0.03 °C [−0.18, 0.22] and 0.10 °C [−0.05, 0.26]). Air regression is 0.13 °C [0.03, 0.22], small but distinguishable; donor regression is −0.09 °C [−0.24, 0.06]. Linear interpolation is the large exception at 1.16 °C [1.00, 1.33]: when the window is entirely unobserved, an interpolator must bridge the whole gap and its error grows

with the bridge, whereas a climatological or donor method substitutes an estimate that does not depend on the length of the missing run.

Figure 3c puts the two findings on one axis. Across the full range of hidden support every reconstructor's error rises monotonically with the share hidden, but the curves are at different heights and the interpolation curve bends upward at the right-hand end. The support share is the right conditioning variable for every method; the coefficient on it, and whether a length term survives alongside it, are properties of the reconstructor. A claim that position dominates length as a law would therefore be wrong, and it is not the claim we make.

4.3 Why the window matters, and what support alone misses

The proximate mechanism is window misalignment. The climatological window that a deployable support rule uses is not the year's true window; in 61% of cases under day-of-year climatology, 58% under smoothing, 56% under interpolation, 37% under donor regression and 51% under air regression, the reconstruction's best 30-day window lands somewhere else than the truth, a median of 4 to 14 days away. When it does, the error is larger by 0.66, 0.65, 0.70, 0.28 and 0.52 °C respectively, and every interval excludes zero (Figure 4a). A rule built on support is therefore a rule built on the climatological guess about where the support is, and it inherits that guess's error.

Two further diagnostics constrain the explanation. Reconstruction residuals are strongly persistent: the median lag-1 autocorrelation of the residual inside the gap is between 0.84 and 0.88 for every method, and cases above a method's median carry 0.16 to 0.36 °C more error under the three climatological and interpolating methods (Figure 4b). Errors therefore do not average out over a gap; they accumulate coherently, which is why a hidden peak costs more than the number of hidden days would suggest. A competing thermal peak does not order the error (Figure 4c): the relation is non-monotone and flat for donor regression, so a close second-best window is not by itself a failure condition. The mechanism is placement, not competition.

Nothing in this section makes the support rule sufficient. Figure 4d shows why: within a single support bin, records refused by the fitted risk score fail at 40 to 55%

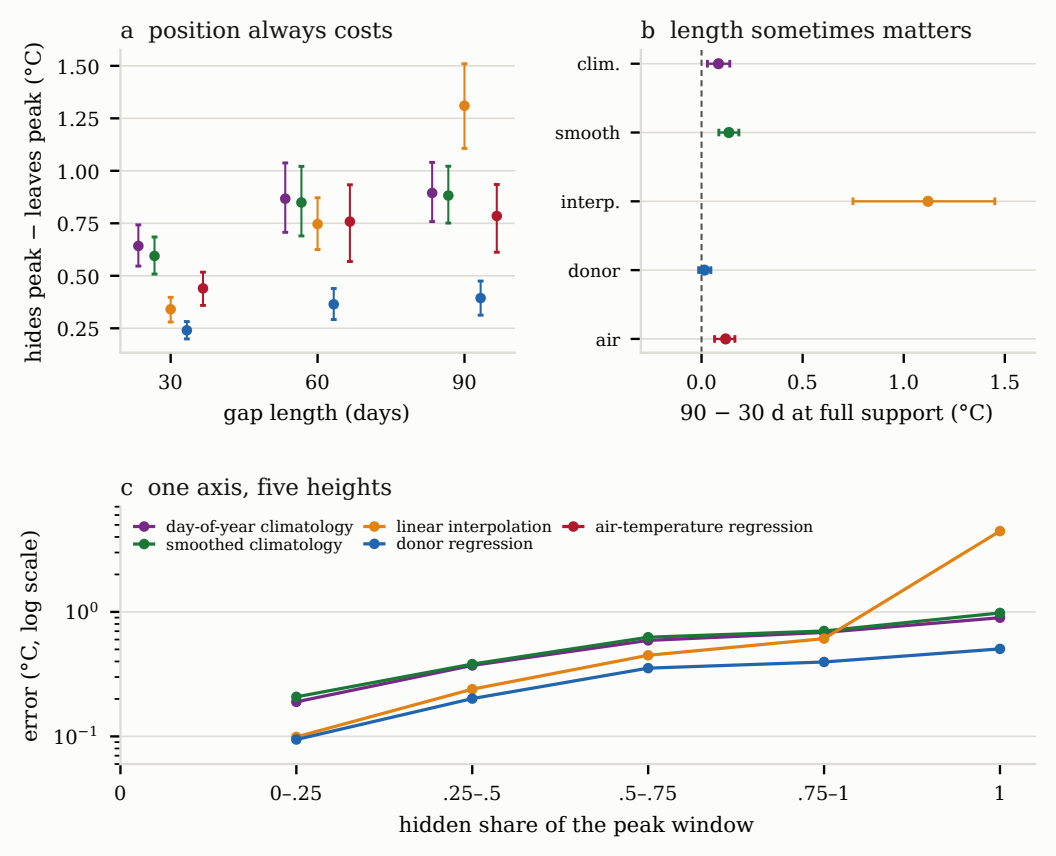


Figure 3. The support effect is universal in sign and conditional in size. (a) Difference in mean absolute error between an outage that hides the whole peak window and one that leaves it observed, at fixed length (30, 60, 90 days), for five reconstructors. (b) Difference between a 90-day and a 30-day outage at a fixed hidden share of one, with a network-paired interval; only linear interpolation retains a length effect once the window is unobserved. (c) Mean absolute error against hidden support share for the same five reconstructors on the external designed panel. Panels (a) and (b) use the crossed panel of 1,899 cases and 12 networks with 2,000 network-clustered bootstrap draws; panel (c) uses the 13,017 external cases with group-clustered intervals. Source data: `fig3_source.csv`.

while records it admits fail at 0.7 to 15%. The score is reading something support does not contain — donor availability, the geometry of scattered missing days, and the reconstructor's own training residual — and the smallest bins are the ones where it matters most. That is the information behind the retention gain reported next, and it is also the reason a more complex rule can be worth its cost where gaps are consequential.

4.4 What it changes for the decision

4.4.1 Retention at the cap, and the comparison that was pre-specified

Figure 5a gives the retention each rule achieves at the pre-registered 5% cap, averaged over the three fold seeds, on the external and development panels. On the external panel the gap-length rule keeps 39.6% of candidates, the peak-window rule 66.7%, the certificate bound 72.9% and the fitted risk score 74.0%. The fitted-score gain over the peak-window rule is 8.0, 7.3 and 7.0 points at seeds 101, 202 and 303, every interval excludes zero, and all three clear the five-point practical margin that was fixed before the comparison (Figure 5d). The development panel reproduces the sign and the size: 70.4% against 60.4%, a gain of 9.7 to 10.3 points. Against the certificate bound the fitted score is not reliably superior: it is +1.6 points [+0.2, +3.1] on the external panel at seed 101 but +1.1 and +0.6 with intervals that include zero at the other two seeds, and −2.6, −1.7 and −2.5 points on development, where the interval includes zero at all three. We therefore report an advantage over a transparent support rule and no reliable advantage over the calibrated bound.

At a fixed 70% retention the ordering is the same and the scale is easier to read: the fitted score admits 2.2% failures on the external panel against 3.7% for the peak-window rule, 8.3% for the gap-length rule and 2.8% for the bound. The gap-length rule is not a straw baseline that a weak comparison flatters: at 70% retention it is the worst of the four by more than five points, which is the cost of asking the wrong question of the archive.

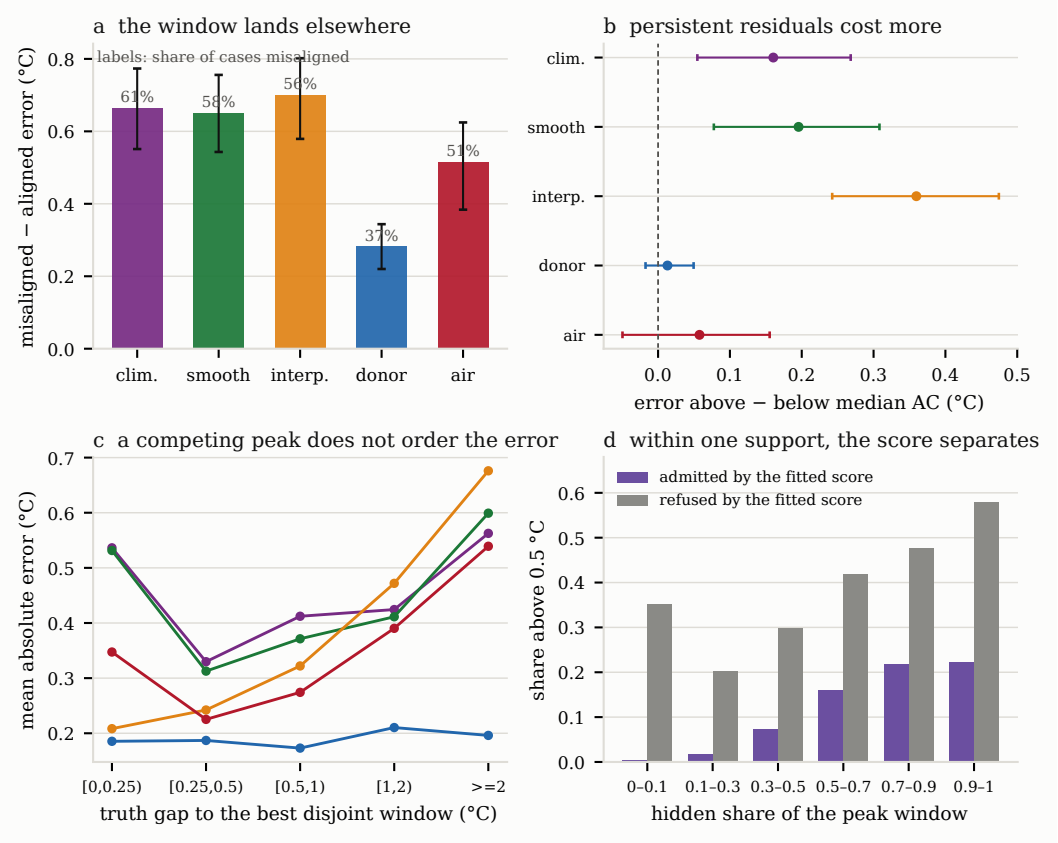


Figure 4. Placement, persistence and the residual information in a fitted score. (a) Increase in mean absolute error when the reconstructed peak window differs from the true one, by reconstructor; labels give the share of cases misaligned. (b) Difference in mean absolute error between cases above and below the reconstructor's median gap-residual lag-1 autocorrelation (median 0.84–0.88 inside the gap). (c) Mean absolute error against the temperature gap to the best disjoint window; a competing peak does not order the error, and the colours follow panels (a) and (b). Panels (a)–(c) use the 1,899-case crossed panel and the network-clustered intervals of the mechanism run. (d) Within each hidden-support bin of the external panel (fold seed 101), the share of records above the 0.5 °C failure threshold among those the fitted score admits and those it refuses. Source data: `fig4_source.csv`.

4.4.2 What refusing a record costs the hot-year archive

Figure 5b is the part of the decision that the designed panel cannot show. On the 418 real calendars, admitting every reconstruction puts 2.6–4.1% of records above the tolerance, under the cap at every point estimate, although the intervals are wide and no cap is certified for a new archive. The screens keep 93–96% of candidates — they remove only a small tail — and reduce the failure rate to 0.5–1.7%. The fitted-score advantage over the peak-window rule shrinks to 0.2–2.4 points, and only 2 of 9 level-by-seed intervals exclude zero. Where the peak window is usually observed, screening has almost nothing to do.

Figure 5c shows what the small tail costs. At a 20 °C hot-year threshold on the external panel, 86% of candidates are true hot years. Reconstructing everything refuses none of them but misjudges 1.3% as cool. The fitted score refuses 25% of the true hot years, the certificate bound 26%, the peak-window rule 33% and the gap-length rule 60%. The admitted hot-year share rises by 1.1 points under the fitted score, for a total deviation of +0.7 points with a 95% interval of −0.3 to +1.6, so the screens do not distort the archive's composition much, but they buy their lower error rate with a large refusal of the years the analysis exists to count. That trade is the decision an agency actually faces, and it is invisible in a table that reports only the error rate among admitted records.

4.5 The risk rule delivers the risk, and the earlier number did not

Two properties of the selector need to be separated because an earlier version of this analysis conflated them. The first is the population risk of the threshold the rule selects: if the archive were infinite, would the admitted set violate the 5% cap? The second is the error rate observed in a finite fresh batch, which carries its own sampling noise. We re-ran the selector on 2,000 synthetic clustered archives per scenario, keeping the deployed rule identical to the one used on the river panels, and measured both.

With the deployed simultaneous bound, the selected threshold's *population* failure rate exceeds 5% in 0.4% of repetitions in the baseline scenario, 0.6% near the cap, 0.4% with unequal group sizes and 2.4% with only twelve calibration groups. The finite fresh batch exceeds 5% far more often — 7.8%, 7.5%, 4.6% and 12.2%

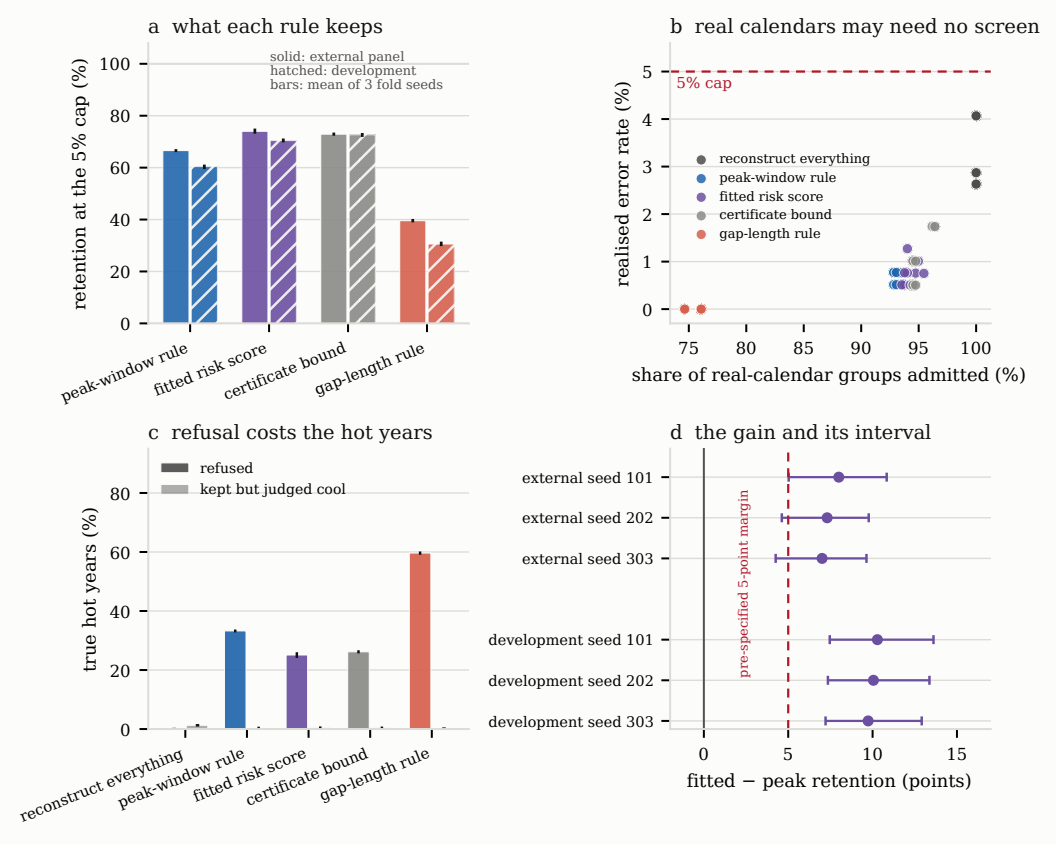


Figure 5. What each rule keeps, and what the keeping costs. (a) Retention at the 5% cap of the admitted reconstruction's error, by rule, on the external panel (solid) and the development panel (hatched); bars are the mean of three fold seeds and the line spans the seeds. The reference arm that admits every reconstruction is omitted here because it does not satisfy the cap on this panel; it appears in panel (b). (b) Coverage against realised failure rate on the 418 real calendars, one point per co-failure level and fold seed, for all five arms; the dashed line is the 5% cap. (c) At a 20 °C hot-year threshold on the external panel, the share of true hot years refused by each rule (dark) and the share admitted but judged cool (light); bars span the three fold seeds. (d) Fitted-score minus peak-window retention with its group-clustered 95% interval, by fold seed, for both panels; the dashed line is the pre-specified five-point margin. Source data: `fig5_source.csv`.

in the same scenarios — because the batch adds noise that the rule cannot control. The 8.5% figure quoted in the previous version of this work was the second quantity, and reading it as the selector's error rate was a misinterpretation; the rule is conservative, not miscalibrated. The price is retention: the simultaneous bound keeps 18.5% of synthetic candidates where a pointwise bound keeps 35.9%.

The extreme percentile the rule reads, $1 - 0.05/86 \approx 0.9994$, is numerically delicate at 1,000 bootstrap draws. Re-running the selection with 10,000 draws changes the chosen threshold in 42 of 150 fold-by-score cells. The decision consequences are small for the score this paper recommends — the fitted score's retention moves by at most 1.0 point — and larger for the discrete rules, where one grid step can move 3.9 points of retention. We therefore describe the selected threshold as a calibration decision at this granularity, not as a stable optimum.

4.6 What the repair changed, and what it did not

The previous version of this analysis declared the policy reconstruction as the admitted object but calibrated the failure event on a different reconstruction. On the external panel the two failure definitions disagree on 898 of 13,017 cases — 6.9% — and the mismatch mattered: with the event measured on the certificate, the fitted score's gain over the peak-window rule on the external panel was 2.7 points; measured on the admitted reconstruction it is 7.2 points under the same tie rule. Most of the change (about 4.5 points) comes from measuring the right object, and about 1.3 points from admitting whole tie blocks instead of jittering the scores. Neither repair was optional: a rule whose stated object is the policy reconstruction cannot be calibrated on a different one, and a rule whose calibration admits half of a tie block and whose deployment admits all of it is two rules.

5 Discussion

5.1 What is new relative to gap-length guidance

The standing guidance is not wrong about what it was derived from. On the estimands for which it was derived — annual means and annual signal parameters that integrate over the year — we reproduce its behaviour: the error grows with the amount of record removed and the position of the removal is second order. Our dis-

agreement is about the transfer of that result to a named extreme. For the hottest 30-day mean the ordering inverts, and the inversion is not small: at the same gap length, a badly placed 30-day outage costs about twenty times a well-placed one, while a 180-day outage that leaves the peak window observed costs almost nothing. A completeness rule cannot see this because the quantity that matters — the share of the statistic's support that is absent — is not a function of the number of missing days alone.

This also changes what a recovery experiment has to control. A comparison across real gaps cannot identify the support effect, because length and support are entangled in exactly the records where the decision is made. The factorial placement used here is not a cosmetic robustness check; it is what makes the claim testable at all. And the claim it supports is conditional: the support share is the right conditioning variable for every reconstructor we tested, but the coefficient on it, and whether a residual length effect survives, are properties of the reconstructor rather than of rivers.

5.2 Why the fitted score adds seven to ten points, and why that is not a licence

The gain over the peak-window rule is real and large enough to matter for an archive. It comes from two places that the mechanism analysis makes visible. First, within a support bin, the fitted score separates records whose reconstruction will fail from those whose will not, using donor availability, the scattered-gap geometry and the reconstructor's own training residual; the refused tail fails at 40 to 55% against 0.7 to 15% for the admitted bulk. Second, the climatological support rule uses a guessed window, and the score is free to learn when that guess is likely to be wrong.

Neither place justifies a general claim of superiority, and the comparison with the certificate bound says so directly. On the development panel the bound keeps more records at the same risk; on the external panel the difference is at most 1.6 points and crosses zero at two of the three seeds. The honest reading is that a fitted score is one way to recover the information a support rule leaves on the table, not the only way and not demonstrably the best way. What the support rule does better

than either is explain itself: its decision is a statement about the record, and an agency can audit it without the model.

5.3 The regime boundary: when screening cannot pay

The real-calendar result is the one that should govern adoption. The gaps that monitoring programmes actually leave are short and mostly off-peak: a median of seven days, a median of 2.2% of the year missing, and 5.7% of records hiding more than half of the peak window. In that regime the unscreened archive sits under the cap at the point estimate, every screen removes only a small tail, and the largest difference between the fitted score and the transparent rule is 2.4 points. The practical question there is not which screen to use but whether to screen at all, and for a 5% tolerance the answer is often no.

Screening earns its cost in the other regime: long gaps, at or over the peak window, which is where the designed panel lives, the fitted score's advantage over the support rule is 7 to 8 points, and the support rule's advantage over the length rule is about 27. An agency can decide which regime it is in before looking at any label, because the gap-length distribution and the climatological overlap are both observable. That is the operational content of this paper: not "screen on support", but "compute the support share first, and if the archive rarely hides the peak, admit the records and spend the effort elsewhere".

The hot-year composition makes the same point from the other side. A screen that lowers the failure rate among admitted records while refusing 25 to 60% of the true hot years has not preserved the archive's purpose, even when the signed composition bias is near zero. The two quantities have to be reported together; one without the other is a number that can always be made to look good.

5.4 Limitations

The external panel is already opened. Every comparison on it is cross-fitted within that panel, not a sealed confirmation on data that took no part in method selection, and the development panel is a sanity replication rather than an independent test. The real-calendar replay is a development-period probe built from

418 candidate groups in 14 networks; its intervals are wide and its point estimates should not be extrapolated to a national archive.

The tolerance of 0.5 °C is a judgement, not a measurement. It corresponds to about half the within-station interannual standard deviation of the hottest 30-day mean in this panel, but an agency with a different use — a criterion exceedance, say — would set a different one, and the retention numbers scale with it.

The record's measurement basis is not uniform. About 74% of development cases are provider-declared daily means, 15% are project-aggregated sub-daily data, 8% have an unspecified daily definition, and 2% are morning reference readings; on the external panel about 94% are declared daily means. We re-selected the thresholds inside the declared-daily-mean subset and the ranking does not depend on the mixed-measurement providers: the fitted score's advantage over the peak-window rule is 8.0, 7.3 and 7.3 points on the external subset and 8.8, 8.7 and 10.4 on the development subset, close to the full-panel values, and its comparison with the certificate bound remains interval-crossing.

Finally, the support share is defined against the year's own peak window, which is unknown at decision time. The deployable rule uses the climatological window and the mechanism section shows what that substitution costs; a rule that could predict the year's window from antecedent conditions would do better, and we have not built one.

6 Conclusions

A completeness rule assumes that the amount of a record that is missing decides whether the record can be used. We tested that assumption by separating what real records entangle: 5,114 complete station-years, 53,721 designed outages that vary gap length and hidden support independently, and 418 real monitoring calendars.

We found that the share of the target statistic's own support that a gap hides orders the error of an annual thermal extreme, whereas the amount missing orders the error of an annual mean; the two orderings are opposite, and the same 30-day gap costs 0.68 °C when it hides the peak window and 0.03 °C when it does not. The support effect is conditional in size: it ranges from 0.24 to 0.64 °C at 30 days across

five reconstructors, and only linear interpolation retains a large length effect once the window is unobserved. The proximate mechanism is window placement — when the reconstructed peak window lands elsewhere the error rises by 0.28 to 0.70 °C — reinforced by residuals that are strongly persistent and therefore do not cancel.

For the admission decision, conditioning on support keeps 67% of an external panel at a 5% error cap where a gap-length rule keeps 40%, and a fitted risk score keeps 74%, 7 to 8 points more than the transparent support rule and not reliably more than a calibrated certificate bound. On the real calendars that agencies actually hold, the unscreened archive sits under the cap at the point estimate and the screens refuse 25 to 60% of the true hot years to remove a small tail. Support is the right conditioning variable; its value is confined to archives whose gaps are consequential, and inside that regime a transparent support rule captures most of what a complex one does.

Open Research Statement

All data are public monitoring records reused under their providers' terms, and the derived tables are regenerated by the scripts listed below. The repository holds the panel-building code, the analysis scripts, the frozen thresholds and the figure sources; the schematic figures are draw.io sources under `figures/wrr/drawio/`, exported to PDF with draw.io 32.5.5, and every data figure is a Python script writing the PDF that appears here, with the table behind each panel in `results/wrr_submission_ready/major_revision_20261020/p0_06_figures/`; the scripts that produce every number in this manuscript are `scripts/330` to `scripts/336`; `scripts/337` exports the schematics and `scripts/338` re-checks the bibliography, with the earlier evidence chain in `scripts/301` to `scripts/324`. Inputs, output hashes and software versions are recorded in each run's `protocol_fingerprint.json`. A permanent archive with a DOI is planned at acceptance; the current repository is a development repository and is not offered as the archival copy. Provider licences and the processing steps that produce the daily series are documented in `DATA_RIGHTS.md` and the supporting information.

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